IBM DATA SCIENCE PROFESSIONAL CAPSTONE

SpaceX Falcon 9 First Stage Landing Prediction

End-to-End Machine Learning, Geospatial & SQL Analytics Capstone

Domain: Aerospace Engineering & Commercial Rocketry Economics

Methodologies: REST API Collection, Web Scraping, Data Wrangling, EDA, SQLite BI, Folium GIS, Plotly Dash, Scikit-Learn Classifiers

Prepared For: IBM Data Science Capstone Final Evaluation

EXECUTIVE SUMMARY

Project Overview & Core Results

• **Business Problem:** SpaceX achieves industry-low launch pricing ($62M vs $165M competitors) through first-stage booster reuse.

• **Data Pipeline:** 90 historical Falcon 9 launches collected, cleaned, and engineered with an empirical 66.7% landing success rate.

• **EDA Insights:** Demonstrated an operational learning curve from 0% early success to >85% in recent launch years. KSC LC 39A achieved 77.3% success.

• **SQL & GIS:** Executed 12 SQL queries in SQLite; mapped launchpads and verified ocean-facing safety corridors via Folium.

• **Machine Learning:** Tuned Logistic Regression, SVM, Decision Trees, and KNN. Best models achieved **83.3% test accuracy** and **0.889 F1-score**.

BUSINESS PROBLEM

Commercial Launch Economics & Opportunity

- Traditional expendable rocketry discards the entire first-stage booster into the ocean.
- Falcon 9's reusable booster accounts for ~70% of rocket construction costs ($30M-$40M).
- Predicting first stage recovery enables satellite operators and launch aggregators to forecast bidding margins.
- When SpaceX recovers a core, aggressive discounts follow; expended boosters create competitive bidding windows.

DATA COLLECTION

REST API & Web Scraping Pipeline

- **SpaceX REST API v4:** Queried past launches endpoint, unpacking payload masses, launch sites, and landing outcomes.
- **Wikipedia Web Scraping:** Extracted 321 tabular flight records using BeautifulSoup.
- **Fault Tolerance:** Cached raw records locally in data/raw/ with automatic fallback to verified benchmark data.

DATA WRANGLING

Data Cleaning & Target Variable Definition

- **Missing Values:** Replaced missing PayloadMass entries with dataset mean (6,105 kg).
- **Target Variable (Class):** Assigned Class = 1 for successful touchdowns (True ASDS, True RTLS, True Ocean).
- **Target Distribution:** 60 Successful Landings (66.7%) vs 30 Failed / Expended (33.3%).
- **Feature Encoding:** One-hot encoded categorical columns (Orbit, LaunchSite, LandingPad, Serial) yielding 80 predictor features.

EXPLORATORY ANALYSIS

Hypotheses & EDA Framework

- Evaluated temporal launch frequency and booster survival rates from 2010 to 2020.
- Compared launch complex effectiveness across Cape Canaveral, Kennedy Space Center, and Vandenberg.
- Investigated orbit destination difficulty (LEO/ISS vs heavy geostationary transfer orbit GTO).

EDA VISUAL RESULTS

Launch Cadence & Learning Curve

- Launch cadence grew exponentially from 1 launch in 2010 to ~19 launches in 2020.
- Success rate followed a sharp learning curve, reaching 80-90% reliability in modern Block 5 flights.
- High resolution charts generated and saved in visuals/eda/.

EDA ORBIT & PAYLOAD

Payload Thresholds & Orbit Feasibility

- Orbits with 100% landing success: ES-L1, GEO, HEO, and SSO.
- Payloads between 2,000 kg and 6,000 kg demonstrate optimal drone ship recovery success.
- Heavy GTO payloads push rocket velocity limits, requiring high-stress ocean drone ship recoveries.

SQL ANALYSIS

Launch Complex & Payload Intelligence

- Cape Canaveral (CCAFS SLC 40) launched 55 missions and lifted over 300,000 kg total payload.
- Kennedy Space Center (KSC LC 39A) achieved the highest success rate at 77.3%.
- Average payload lifted across all missions is 6,105 kg (range: 350 kg to 15,600 kg).

SQL OUTCOMES

Milestone Achievements & Outcome Rankings

- First successful ground pad landing: 2015-12-22 by Booster B1019 at CCAFS SLC 40.
- Maximum payload missions: 15,600 kg Starlink deployment flights.
- Autonomous Spaceport Drone Ship (True ASDS) is the most frequent success mode (41 recoveries).

GEOSPATIAL INTELLIGENCE

Interactive Folium Launch Map

- Launchpads are sited 0.9 to 1.5 km from ocean shorelines to ensure eastward trajectories over water.
- Proximity to highways and railways enables heavy booster transport while preserving 10-30 km buffer zones to cities.
- Interactive map saved with MarkerClusters and coordinate telemetry in visuals/folium/launch_sites_map.html.

INTERACTIVE ANALYTICS

Plotly Dash Operations Dashboard

- Standalone web application in dashboard/app.py with dark aerospace UI.
- Dynamic filters: Launch Site dropdown, Outcome radio toggle, and Payload range slider.
- Reactive KPI summary cards, pie charts, scatter correlations, and yearly cadence bars.

PREDICTIVE METHODOLOGY

Machine Learning Pipeline & Leakage Prevention

- 80 independent features including core reuse count, grid fins, landing legs, and orbital profiles.
- Strict leakage prevention: StandardScaler fitted strictly on 80% training partition.
- 10-Fold Stratified Cross-Validation combined with GridSearchCV for hyperparameter optimization.

MODEL COMPARISON

Supervised Classification Performance

- **K-Nearest Neighbors & Logistic Regression:** Achieved **83.3% test accuracy** and **0.889 F1-score**.
- **Support Vector Machine & Decision Tree:** Achieved 77.8% test accuracy and ~0.85 F1-score.
- Both top models correctly classified 11 out of 12 successful landings on held-out test data.

CONCLUSIONS & IMPACT

Commercial Insights & Future Outlook

- First-stage reuse provides SpaceX an estimated $40M cost advantage per flight.
- Machine learning provides viable predictive power to forecast booster recovery before flight.
- Future Work: Integrate ocean wave height and drone ship sea-state telemetry for entry burn modeling.

SUBMISSION & REFERENCES

Repository Structure & Submission Details

- Complete runnable codebase with notebooks, scripts, database, dashboard, and visuals.
- Sources: SpaceX REST API v4, Wikipedia Launch History, IBM Data Science Professional Certificate.
- GitHub Repository: Ready for push and grading.